

Beginning with ONTAP 9.9.0:



- ONTAP 9.x.0 releases are cloud-only releases and support Cloud Volumes ONTAP (CVO) systems. The asterisk (*) after the release version indicates a cloud-only release.
- ONTAP 9.x.1 releases are general releases and support both on-premises and CVO systems.



Locate the higher, more recent ONTAP version in the left column, and in the top row locate the lower ONTAP version to determine interoperability. Interoperability is bidirectional.

Interoperability for ONTAP version 9.3 and later

ONTAP version...	Interoperates with these previous ONTAP versions...															
	9.13.1	9.13.0*	9.12.1	9.12.0*	9.11.1	9.11.0*	9.10.1	9.10.0*	9.9.1	9.9.0*	9.8	9.7	9.6	9.5	9.4	9.3
9.13.1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No
9.13.0*	Yes	Yes	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	No	No	No	No
9.12.1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No
9.12.0*	Yes	No	Yes	Yes	Yes	No	Yes	No	Yes	No	Yes	Yes	No	No	No	No
9.11.1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No
9.11.0*	Yes	No	Yes	No	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes	No	No	No
9.10.1	Yes	Yes	Yes	Yes	n/a	n/a	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
9.10.0*	Yes	No	Yes	No	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No
9.9.1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
9.9.0*	Yes	No	Yes	No	Yes	No	Yes	n/a	Yes	Yes	Yes	Yes	Yes	Yes	No	No
9.8	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
9.7	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
9.6	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes
9.5	No	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
9.4	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes

9.3	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes	Yes
-----	----	----	----	----	----	----	----	----	----	----	----	-----	-----	-----	-----	-----	-----

SnapMirror Synchronous relationships



SnapMirror Synchronous is not supported for ONTAP cloud instances.

ONTAP version ...	Interoperates with these previous ONTAP versions...								
	9.13.1	9.12.1	9.11.1	9.10.1	9.9.1	9.8	9.7	9.6	9.5
9.13.1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
9.12.1	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No
9.11.1	Yes	Yes	Yes	Yes	Yes	No	No	No	No
9.10.1	No	Yes	Yes	Yes	Yes	Yes	No	No	No
9.9.1	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No
9.8	No	Yes	No	Yes	Yes	Yes	Yes	Yes	No
9.7	No	Yes	No	No	Yes	Yes	Yes	Yes	Yes
9.6	No	No	No	No	No	Yes	Yes	Yes	Yes
9.5	No	No	No	No	No	No	Yes	Yes	Yes

SnapMirror SVM DR relationships

For SVM DR data and SVM protection:

SVM DR is only supported between clusters running the same version of ONTAP.

For SVM DR for SVM migration:

- Replication is supported in a single direction from an earlier version of ONTAP to a later version of ONTAP; for example, from ONTAP 9.11.1 to ONTAP 9.12.1
- The ONTAP version on the target cluster must be no more than 2 versions newer, as shown in the table below.
- Replication is not supported for long-term data protection use cases.

The asterisk (*) after the release version indicates a cloud-only release.

Sour ce	Destination															
	9.3	9.4	9.5	9.6	9.7	9.8	9.9.0 *	9.9.1	9.10.0 *	9.10.1	9.11.0 *	9.11.1	9.12.0 *	9.12.1	9.13.0 *	9.13.1
9.3	Yes	Yes	Yes													
9.4		Yes	Yes	Yes												
9.5			Yes	Yes	Yes											

9.6				Yes	Yes	Yes										
9.7					Yes	Yes	Yes									
9.8						Yes	Yes	Yes								
9.9.0*							Yes	Yes	Yes							
9.9.1								Yes	Yes	Yes						
9.10.0*									Yes	Yes	Yes					
9.10.1										Yes	Yes	Yes				
9.11.0*											Yes	Yes	Yes			
9.11.1												Yes	Yes	Yes		
9.12.0*													Yes	Yes	Yes	
9.12.1														Yes	Yes	Yes
9.13.0*															Yes	Yes
9.13.1																Yes

SnapMirror DR relationships

For SnapMirror relationships of type “DP” and policy type “async-mirror”:



DP-type mirrors cannot be initialized beginning with ONTAP 9.11.1 and are completely deprecated in ONTAP 9.12.1. For more information, see [Deprecation of data protection SnapMirror relationships](#).



In the following table, the column on the left indicates the ONTAP version on the source volume, and the top row indicates the ONTAP versions you can have on your destination volume.

Source	Destination											
	9.11.1	9.10.1	9.9.1	9.8	9.7	9.6	9.5	9.4	9.3	9.2	9.1	9
9.11.1	Yes	No	No	No	No	No	No	No	No	No	No	No
9.10.1	Yes	Yes	No	No	No	No	No	No	No	No	No	No
9.9.1	Yes	Yes	Yes	No	No	No	No	No	No	No	No	No
9.8	No	Yes	Yes	Yes	No	No	No	No	No	No	No	No
9.7	No	No	Yes	Yes	Yes	No	No	No	No	No	No	No

9.6	No	No	No	Yes	Yes	Yes	No	No	No	No	No	No
9.5	No	No	No	No	Yes	Yes	Yes	No	No	No	No	No
9.4	No	No	No	No	No	Yes	Yes	Yes	No	No	No	No
9.3	No	No	No	No	No	No	Yes	Yes	Yes	No	No	No
9.2	No	No	No	No	No	No	No	Yes	Yes	Yes	No	No
9.1	No	No	No	No	No	No	No	No	Yes	Yes	Yes	No
9	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes



Interoperability is not bidirectional.

SnapMirror limitations

You should be aware of basic SnapMirror limitations before creating a data protection relationship.

- A destination volume can have only one source volume.



A source volume can have multiple destination volumes. The destination volume can be the source volume for any type of SnapMirror replication relationship.

- Depending on the array model, you can fan out a maximum of eight or sixteen destination volumes from a single source volume. See the [Hardware Universe](#) to learn details for your specific configuration.
- You cannot restore files to the destination of a SnapMirror DR relationship.
- Source or destination SnapVault volumes cannot be 32-bit.
- The source volume for a SnapVault relationship should not be a FlexClone volume.



The relationship will work, but the efficiency offered by FlexClone volumes will not be preserved.

Archive and compliance using SnapLock technology

What SnapLock is

SnapLock is a high-performance compliance solution for organizations that use WORM storage to retain files in unmodified form for regulatory and governance purposes.

SnapLock helps to prevent deletion, change, or renaming of data to meet regulations such as SEC 17a-4, HIPAA, FINRA, CFTC, and GDPR. With SnapLock, you can create special-purpose volumes in which files can be stored and committed to a non-erasable, non-writable state either for a designated retention period or indefinitely. SnapLock allows this retention to be performed at the file level through standard open file protocols such as CIFS and NFS. The supported open file protocols for SnapLock are NFS (versions 2, 3, and 4) and CIFS (SMB 1.0, 2.0, and 3.0).

Using SnapLock, you commit files and Snapshot copies to WORM storage, and set retention periods for WORM-protected data. SnapLock WORM storage uses NetApp Snapshot technology and can leverage

SnapMirror replication, and SnapVault backups as the base technology for providing backup recovery protection for data.

Learn more about WORM storage: [Compliant WORM storage using NetApp SnapLock - TR-4526](#).

You can use an application to commit files to WORM over NFS or CIFS, or use the SnapLock autocommit feature to commit files to WORM automatically. You can use a *WORM appendable file* to retain data that is written incrementally, like log information. For more information see [Use volume append mode to create WORM appendable files](#).

SnapLock supports data protection methods that should satisfy most compliance requirements:

- You can use SnapLock for SnapVault to WORM-protect Snapshot copies on secondary storage. See [Commit Snapshot copies to WORM](#).
- You can use SnapMirror to replicate WORM files to another geographic location for disaster recovery. See [Mirror WORM files](#).

SnapLock is a license-based feature of NetApp ONTAP. A single license entitles you to use SnapLock in strict Compliance mode, to satisfy external mandates like SEC Rule 17a-4, and a looser Enterprise mode, to meet internally mandated regulations for the protection of digital assets. SnapLock licenses are part of the Security and Compliance bundle.

SnapLock is supported on all AFF and FAS systems as well as ONTAP Select. SnapLock is not a software-only solution; it is an integrated hardware and software solution. This distinction is important for strict WORM regulations such as SEC 17a-4, which requires an integrated hardware and software solution. For more information, refer to [SEC Interpretation: Electronic Storage of Broker-Dealer Records](#).

What you can do with SnapLock

After you configure SnapLock, you can complete the following tasks:

- [Commit files to WORM](#)
- [Commit Snapshot copies to WORM for secondary storage](#)
- [Mirror WORM files for disaster recovery](#)
- [Retain WORM files during litigation using Legal Hold](#)
- [Delete WORM files using the privileged delete feature](#)
- [Set the file retention period](#)
- [Move a SnapLock volume](#)
- [Lock a Snapshot copy for protection against ransomware attacks](#)
- [Review SnapLock use with the Audit Log](#)
- [Use SnapLock APIs](#)

SnapLock Compliance and Enterprise modes

SnapLock Compliance and Enterprise modes differ mainly in the level at which each mode protects WORM files:

SnapLock mode	Protection level	WORM file deleting during retention
---------------	------------------	-------------------------------------

Compliance mode	At the file level	Cannot be deleted
Enterprise mode	At the disk level	Can be deleted by the compliance administrator using an audited “privileged delete” procedure

After the retention period has elapsed, you are responsible for deleting any files you no longer need. Once a file has been committed to WORM, whether under Compliance or Enterprise mode, it cannot be modified, even after the retention period has expired.

You cannot move a WORM file during or after the retention period. You can copy a WORM file, but the copy will not retain its WORM characteristics.

The following table shows the differences in capabilities supported by SnapLock Compliance and Enterprise modes:

Capability	SnapLock Compliance	SnapLock Enterprise
Enable and delete files using privileged delete	No	Yes
Reinitialize disks	No	Yes
Destroy SnapLock aggregates and volumes during retention period	No	Yes, with the exception of the SnapLock audit log volume
Rename aggregates or volumes	No	Yes
Use non-NetApp disks	No	Yes (with FlexArray Virtualization)
Use the SnapLock volume for audit logging	Yes	Yes, beginning with ONTAP 9.5

Supported and unsupported features with SnapLock

The following table shows the features that are supported with SnapLock Compliance mode, SnapLock Enterprise mode, or both:

Feature	Supported with SnapLock Compliance	Supported with SnapLock Enterprise
Consistency Groups	No	No
Encrypted volumes	Yes, beginning with ONTAP 9.2. Learn more about Encryption and SnapLock .	Yes, beginning with ONTAP 9.2. Learn more about Encryption and SnapLock .

FabricPools on SnapLock aggregates	No	Yes, beginning with ONTAP 9.8. Learn more about FabricPool on SnapLock Enterprise aggregates .
Flash Pool aggregates	Yes, beginning with ONTAP 9.1.	Yes, beginning with ONTAP 9.1.
FlexClone	You can clone SnapLock volumes, but you cannot clone files on a SnapLock volume.	You can clone SnapLock volumes, but you cannot clone files on a SnapLock volume.
FlexGroup volumes	Yes, beginning with ONTAP 9.11.1. Learn more about FlexGroup volumes .	Yes, beginning with ONTAP 9.11.1. Learn more about FlexGroup volumes .
LUNs	No	No
MetroCluster configurations	Yes, beginning with ONTAP 9.3. Learn more about MetroCluster support .	Yes, beginning with ONTAP 9.3. Learn more about MetroCluster support .
Multi-admin verification (MAV)	Yes, beginning with ONTAP 9.13.1. Learn more about MAV support .	Yes, beginning with ONTAP 9.13.1. Learn more about MAV support .
SAN	No	No
Single-file SnapRestore	No	Yes
SnapMirror Business Continuity	No	No
SnapRestore	No	Yes
SMTape	No	No
SnapMirror Synchronous	No	No
SSDs	Yes, beginning with ONTAP 9.1.	Yes, beginning with ONTAP 9.1.
Storage efficiency features	Yes, beginning with ONTAP 9.9.1. Learn more about storage efficiency support .	Yes, beginning with ONTAP 9.9.1. Learn more about storage efficiency support .

FabricPool on SnapLock Enterprise aggregates

FabricPools are supported on SnapLock Enterprise aggregates beginning with ONTAP 9.8. However, your account team needs to open a product variance request documenting that you understand that FabricPool data tiered to a public or private cloud is no longer protected by SnapLock because a cloud admin can delete that data.



Any data that FabricPool tiers to a public or private cloud is no longer protected by SnapLock because that data can be deleted by a cloud administrator.

FlexGroup volumes

SnapLock supports FlexGroup volumes beginning with ONTAP 9.11.1; however, the following features are not supported:

- Legal-hold
- Event-based retention
- SnapLock for SnapVault (supported beginning with ONTAP 9.12.1)

You should also be aware of the following behaviors:

- The volume compliance clock (VCC) of a FlexGroup volume is determined by the VCC of the root constituent. All non-root constituents will have their VCC closely synced to the root VCC.
- SnapLock configuration properties are set only on the FlexGroup as a whole. Individual constituents cannot have different configuration properties, such as default retention time and autocommit period.

MetroCluster support

SnapLock support in MetroCluster configurations differs between SnapLock Compliance mode and SnapLock Enterprise mode.

SnapLock Compliance

- Beginning with ONTAP 9.3, SnapLock Compliance is supported on unmirrored MetroCluster aggregates.
- Beginning with ONTAP 9.3, SnapLock Compliance is supported on mirrored aggregates, but only if the aggregate is used to host SnapLock audit log volumes.
- SVM-specific SnapLock configurations can be replicated to primary and secondary sites using MetroCluster.

SnapLock Enterprise

- Beginning with ONTAP 9, SnapLock Enterprise aggregates are supported.
- Beginning with ONTAP 9.3, SnapLock Enterprise aggregates with privileged delete are supported.
- SVM-specific SnapLock configurations can be replicated to both sites using MetroCluster.

MetroCluster configurations and compliance clocks

MetroCluster configurations use two compliance clock mechanisms, the Volume Compliance Clock (VCC) and the System Compliance Clock (SCC). The VCC and SCC are available to all SnapLock configurations. When you create a new volume on a node, its VCC is initialized with the current value of the SCC on that node. After the volume is created, the volume and file retention time is always tracked with the VCC.

When a volume is replicated to another site, its VCC is also replicated. When a volume switchover occurs, from Site A to Site B, for example, the VCC continues to be updated on Site B while the SCC on Site A halts when Site A goes offline.

When Site A is brought back online and the volume switchback is performed, the Site A SCC clock restarts while the VCC of the volume continues to be updated. Because the VCC is continuously updated, regardless of switchover and switchback operations, the file retention times do not depend on SCC clocks and do not stretch.

Multi-admin verification (MAV) support

Beginning with ONTAP 9.13.1, a cluster administrator can explicitly enable multi-admin verification on a cluster to require quorum approval before some SnapLock operations are executed. When MAV is enabled, SnapLock volume properties such as default-retention-time, minimum-retention-time, maximum-retention-time, volume-append-mode, autocommit-period and privileged-delete will require quorum approval. Learn more about [MAV](#).

Storage efficiency

Beginning with ONTAP 9.9.1, SnapLock supports storage efficiency features, such as data compaction, cross-volume-deduplication, and adaptive compression for SnapLock volumes and aggregates. For more information about storage efficiency, see [Logical storage management overview with the CLI](#).

Encryption

ONTAP offers both software- and hardware-based encryption technologies for ensuring that data at rest cannot be read if the storage medium is repurposed, returned, misplaced, or stolen.

Disclaimer: NetApp cannot guarantee that SnapLock-protected WORM files on self-encrypting drives or volumes will be retrievable if the authentication key is lost or if the number of failed authentication attempts exceeds the specified limit and results in the drive being permanently locked. You are responsible for ensuring against authentication failures.



Beginning with ONTAP 9.2, encrypted volumes are supported on SnapLock aggregates.

7-Mode Transition

You can migrate SnapLock volumes from 7-Mode to ONTAP by using the Copy-Based Transition (CBT) feature of the 7-Mode Transition Tool. The SnapLock mode of the destination volume, Compliance or Enterprise, must match the SnapLock mode of the source volume. You cannot use Copy-Free Transition (CFT) to migrate SnapLock volumes.

Configure SnapLock

Configure SnapLock

Before you use SnapLock, you need to configure SnapLock by completing various tasks such as install the SnapLock license, initialize Compliance Clock, create a SnapLock aggregate and more.

Install the license

A SnapLock license entitles you to use both SnapLock Compliance mode and SnapLock Enterprise mode. SnapLock licenses are issued on a per-node basis. You must install a license for each node that hosts a SnapLock aggregate.

For details about Compliance mode and Enterprise mode, see [What SnapLock is](#).

What you'll need

You must be a cluster administrator to perform this task.

About this task

You should have received the SnapLock license keys from your sales representative.

Perform this task using ONTAP System Manager or the ONTAP CLI.

System Manager

1. Navigate to **Cluster > Settings > Licenses > Add License**.
2. Click **+Add**.
3. Click **Browse** and locate the NetApp License File.
4. Click **Add**.

CLI

1. Install the SnapLock license for a node:

```
system license add -license-code license_key
```

The following command installs the license with the key `AAAAAAAAAAAAAAAAAAAAAAAAAAAA`.

```
cluster1::> system license add -license-code  
AAAAAAAAAAAAAAAAAAAAAAAAAAAA
```

2. Repeat the previous step for each node license.

Initialize the ComplianceClock

The SnapLock ComplianceClock ensures against tampering that might alter the retention period for WORM files. You must initialize the *system ComplianceClock* on each node that hosts a SnapLock aggregate. Once you initialize the ComplianceClock on a node, you cannot initialize it again.

What you'll need

- You must be a cluster administrator to perform this task.
- The SnapLock license must be installed on the node.

About this task

The time on the system ComplianceClock is inherited by the *volume ComplianceClock*, which controls the retention period for WORM files on the volume. The volume ComplianceClock is initialized automatically when you create a new SnapLock volume.



The initial setting of the ComplianceClock is based on the current system clock. For that reason, you should verify that the system time and time zone are correct before initializing the ComplianceClock. Once you initialize the ComplianceClock on a node, you cannot initialize it again.

System Manager

Beginning with ONTAP 9.12.1, you can use System Manager to initialize the SnapLock Compliance Clock.

Steps

1. Navigate to **Cluster > Overview**.
2. In the **Nodes** section, click **Initialize SnapLock Compliance Clock**.
3. To display the Compliance Clock column and to verify that the Compliance Clock is initialized, in the **Cluster > Overview > Nodes** section, click **Show/Hide** and select **SnapLock Compliance Clock**.

CLI

1. Initialize the system ComplianceClock:

```
snaplock compliance-clock initialize -node node_name
```

The following command initializes the system ComplianceClock on node1:

```
cluster1::> snaplock compliance-clock initialize -node node1
```

2. When prompted, confirm that the system clock is correct and that you want to initialize the ComplianceClock:

```
Warning: You are about to initialize the secure ComplianceClock of
the node "node1" to the current value of the node's system clock.
This procedure can be performed only once on a given node, so you
should ensure that the system time is set correctly before
proceeding.
```

```
The current node's system clock is: Mon Apr 25 06:04:10 GMT 2016
```

```
Do you want to continue? (y|n): y
```

3. Repeat this procedure for each node that hosts a SnapLock aggregate.

Enable ComplianceClock resynchronization for an NTP-configured system

You can enable the SnapLock ComplianceClock time synchronization feature when an NTP server is configured.

What you'll need

- This feature is available only at the advanced privilege level.
- You must be a cluster administrator to perform this task.
- The SnapLock license must be installed on the node.
- This feature is available only for Cloud Volumes ONTAP, ONTAP Select, and VSIM platforms.

About this task

When the SnapLock secure clock daemon detects a skew beyond the threshold, ONTAP uses the system time to reset both the system and volume ComplianceClocks. A period of 24 hours is set as the skew threshold. This means that the system ComplianceClock is synchronized to the system clock only if the skew is more than a day old.

The SnapLock secure clock daemon detects a skew and changes the ComplianceClock to the system time. Any attempt at modifying the system time to force the ComplianceClock to synchronize to the system time fails, since the ComplianceClock synchronizes to the system time only if the system time is synchronized with the NTP time.

Steps

1. Enable the SnapLock ComplianceClock time synchronization feature when an NTP server is configured:

```
snaplock compliance-clock ntp
```

The following command enables the system ComplianceClock time synchronization feature:

```
cluster1::*> snaplock compliance-clock ntp modify -is-sync-enabled true
```

2. When prompted, confirm that the configured NTP servers are trusted and that the communications channel is secure to enable the feature:
3. Check that the feature is enabled:

```
snaplock compliance-clock ntp show
```

The following command checks that the system ComplianceClock time synchronization feature is enabled:

```
cluster1::*> snaplock compliance-clock ntp show  
  
Enable clock sync to NTP system time: true
```

Create a SnapLock aggregate

You use the volume `-snaplock-type` option to specify a Compliance or Enterprise SnapLock volume type. For releases earlier than ONTAP 9.10.1, you must create a separate SnapLock aggregate. Beginning with ONTAP 9.10.1, SnapLock and non-SnapLock volumes can exist on the same aggregate; therefore, you are no longer required to create a separate SnapLock aggregate if you are using ONTAP 9.10.1.

What you'll need

- You must be a cluster administrator to perform this task.
- The SnapLock license must be installed on the node.
- The ComplianceClock on the node must be initialized.
- If you have partitioned the disks as “root”, “data1”, and “data2”, you must ensure that spare disks are available.

Upgrade considerations

When upgrading to ONTAP 9.10.1, existing SnapLock and non-SnapLock aggregates are upgraded to support the existence of both SnapLock and non-SnapLock volumes; however, the existing SnapLock volume attributes are not automatically updated. For example, data-compaction, cross-volume-dedupe, and cross-volume-background-dedupe fields remain unchanged. New SnapLock volumes created on existing aggregates have the same default values as non-SnapLock volumes, and the default values for new volumes and aggregates are platform dependent.

Revert considerations

If you need to revert to an ONTAP version earlier than 9.10.1, you must move all SnapLock Compliance, SnapLock Enterprise, and SnapLock volumes to their own SnapLock aggregates.

About this task

- You cannot create Compliance aggregates for FlexArray LUNs, but SnapLock Compliance aggregates are supported with FlexArray LUNs.
- You cannot create Compliance aggregates with the SyncMirror option.
- You can create mirrored Compliance aggregates in a MetroCluster configuration only if the aggregate is used to host SnapLock audit log volumes.



In a MetroCluster configuration, SnapLock Enterprise is supported on mirrored and unmirrored aggregates. SnapLock Compliance is supported only on unmirrored aggregates.

Steps

1. Create a SnapLock aggregate:

```
storage aggregate create -aggregate aggregate_name -node node_name -diskcount  
number_of_disks -snaplock-type compliance|enterprise
```

The man page for the command contains a complete list of options.

The following command creates a SnapLock Compliance aggregate named `aggr1` with three disks on `node1`:

```
cluster1::> storage aggregate create -aggregate aggr1 -node node1  
-diskcount 3 -snaplock-type compliance
```

Create and mount SnapLock volumes

You must create a SnapLock volume for the files or Snapshot copies that you want to commit to the WORM state. Beginning with ONTAP 9.10.1, any volume you create, regardless of the aggregate type, is created by default as a non-SnapLock volume. You must use the `-snaplock-type` option to explicitly create a SnapLock volume by specifying either Compliance or Enterprise as the SnapLock type. By default, the SnapLock type is set to `non-snaplock`.

What you'll need

- The SnapLock aggregate must be online.

- The SnapLock license must be installed on the node.
- The ComplianceClock on the node must be initialized.

About this task

With the proper SnapLock permissions, you can destroy or rename an Enterprise volume at any time. You cannot destroy a Compliance volume until the retention period has elapsed. You can never rename a Compliance volume.

You can clone SnapLock volumes, but you cannot clone files on a SnapLock volume. The clone volume will be of the same SnapLock type as the parent volume.



LUNs are not supported on SnapLock volumes. Although it is possible to move LUNs onto a SnapLock volume using legacy technology, this is not a supported operation, nor is any other operation involving LUNs on a SnapLock volume.

Perform this task using ONTAP System Manager or the ONTAP CLI.

System Manager

Beginning with ONTAP 9.12.1, you can use System Manager to create a SnapLock volume.

Steps

1. Navigate to **Storage > Volumes** and click **Add**.
2. In the **Add Volume** window, click **More Options**.
3. Enter the new volume information, including the name and size of the volume.
4. Select **Enable SnapLock** and choose the SnapLock type, either Compliance or Enterprise.
5. In the **Auto-Commit Files** section, select **Modified** and enter the amount of time a file should remain unchanged before it is automatically committed. The minimum value is 5 minutes and the maximum value is 10 years.
6. In the **Data Retention** section, select the minimum and maximum retention period.
7. Select the default retention period.
8. Click **Save**.
9. Select the new volume in the **Volumes** page to verify the SnapLock settings.

CLI

1. Create a SnapLock volume:

```
volume create -vserver SVM_name -volume volume_name -aggregate  
aggregate_name -snaplock-type compliance|enterprise
```

For a complete list of options, see the man page for the command. The following options are not available for SnapLock volumes: `-nvfail`, `-atime-update`, `-is-autobalance-eligible`, `-space-mgmt-try-first`, and `vmalign`.

The following command creates a SnapLock Compliance volume named `vol1` on `aggr1` on `vs1`:

```
cluster1::> volume create -vserver vs1 -volume vol1 -aggregate aggr1  
-snaplock-type compliance
```

Mount a SnapLock volume

You can mount a SnapLock volume to a junction path in the SVM namespace for NAS client access.

What you'll need

The SnapLock volume must be online.

About this task

- You can mount a SnapLock volume only under the root of the SVM.
- You cannot mount a regular volume under a SnapLock volume.

Steps

1. Mount a SnapLock volume:

```
volume mount -vserver SVM_name -volume volume_name -junction-path path
```

For a complete list of options, see the man page for the command.

The following command mounts a SnapLock volume named `vol1` to the junction path `/sales` in the `vs1` namespace:

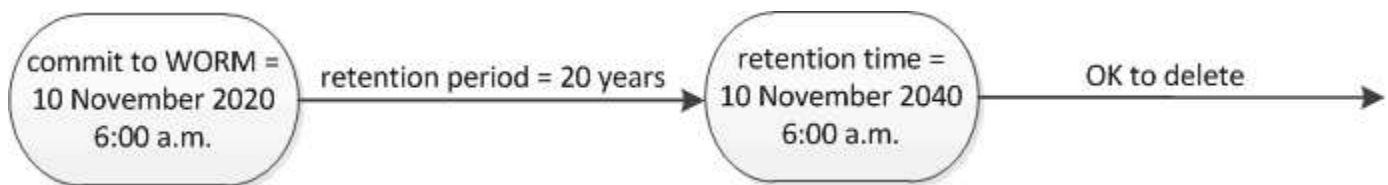
```
cluster1::> volume mount -vserver vs1 -volume vol1 -junction-path /sales
```

Set the retention time

You can set the retention time for a file explicitly, or you can use the default retention period for the volume to derive the retention time. Unless you set the retention time explicitly, SnapLock uses the default retention period to calculate the retention time. You can also set file retention after an event.

About retention period and retention time

The *retention period* for a WORM file specifies the length of time the file must be retained after it is committed to the WORM state. The *retention time* for a WORM file is the time after which the file no longer needs to be retained. A retention period of 20 years for a file committed to the WORM state on 10 November 2020 6:00 a.m., for example, would yield a retention time of 10 November 2040 6:00 a.m.



Beginning with ONTAP 9.10.1, you can set a retention time up to October 26, 3058 and a retention period up to 100 years. When you extend retention dates, older policies are converted automatically. In ONTAP 9.9.1 and earlier releases, unless you set the default retention period to infinite, the maximum supported retention time is January 19 2071 (GMT).

Important replication considerations

When establishing a SnapMirror relationship with a SnapLock source volume using a retention date later than January 19th 2071 (GMT), the destination cluster must be running ONTAP 9.10.1 or later or the SnapMirror transfer will fail.

Important revert considerations

ONTAP prevents you from reverting a cluster from ONTAP 9.10.1 to an earlier ONTAP version when there are any files with a retention period later than “January 19, 2071 8:44:07 AM”.

Understanding the default retention periods

A SnapLock Compliance or Enterprise volume has four retention periods:

- Minimum retention period (`min`), with a default of 0
- Maximum retention period (`max`), with a default of 30 years
- Default retention period, with a default equal to `min` for both Compliance mode and Enterprise mode

beginning with ONTAP 9.10.1. In ONTAP releases earlier than ONTAP 9.10.1, the default retention period depends on the mode:

- For Compliance mode, the default is equal to `max`.
- For Enterprise mode, the default is equal to `min`.
- Unspecified retention period.

Beginning with ONTAP 9.8, you can set the retention period on files in a volume to `unspecified`, to enable the file to be retained until you set an absolute retention time. You can set a file with absolute retention time to unspecified retention and back to absolute retention as long as the new absolute retention time is later than the absolute time you previously set.

Beginning with ONTAP 9.12.1, WORM files with the retention period set to `unspecified` are guaranteed to have a retention period set to the minimum retention period configured for the SnapLock volume. When you change the file retention period from `unspecified` to an absolute retention time, the new retention time specified must be greater than the minimum retention time already set on the file.

So, if you do not set the retention time explicitly before committing a Compliance-mode file to the WORM state, and you do not modify the defaults, the file will be retained for 30 years. Similarly, if you do not set the retention time explicitly before committing an Enterprise-mode file to the WORM state, and you do not modify the defaults, the file will be retained for 0 years, or, effectively, not at all.

Set the default retention period

You can use the `volume snaplock modify` command to set the default retention period for files on a SnapLock volume.

What you'll need

The SnapLock volume must be online.

About this task

The following table shows the possible values for the default retention period option:



The default retention period must be greater than or equal to ($\geq$) the minimum retention period and less than or equal to ($\leq$) the maximum retention period.

Value	Unit	Notes
0 - 65535	seconds	
0 - 24	hours	
0 - 365	days	
0 - 12	months	
0 - 100	years	Beginning with ONTAP 9.10.1. For earlier ONTAP releases, the value is 0 - 70.

Value	Unit	Notes
max	-	Use the maximum retention period.
min	-	Use the minimum retention period.
infinite	-	Retain the files forever.
unspecified	-	Retain the files until an absolute retention period is set.

The values and ranges for the maximum and minimum retention periods are identical, except for `max` and `min`, which are not applicable. For more information about this task, see [Set the retention time overview](#).

You can use the `volume snaplock show` command to view the retention period settings for the volume. For more information, see the man page for the command.



After a file has been committed to the WORM state, you can extend but not shorten the retention period.

Steps

1. Set the default retention period for files on a SnapLock volume:

```
volume snaplock modify -vserver SVM_name -volume volume_name -default  
-retention-period default_retention_period -minimum-retention-period  
min_retention_period -maximum-retention-period max_retention_period
```

For a complete list of options, see the man page for the command.



The following examples assume that the minimum and maximum retention periods have not been modified previously.

The following command sets the default retention period for a Compliance or Enterprise volume to 20 days:

```
cluster1::> volume snaplock modify -vserver vs1 -volume vol1 -default  
-retention-period 20days
```

The following command sets the default retention period for a Compliance volume to 70 years:

```
cluster1::> volume snaplock modify -vserver vs1 -volume vol1 -maximum  
-retention-period 70years
```

The following command sets the default retention period for an Enterprise volume to 10 years:

```
cluster1::> volume snaplock modify -vserver vs1 -volume voll -default  
-retention-period max -maximum-retention-period 10years
```

The following commands set the default retention period for an Enterprise volume to 10 days:

```
cluster1::> volume snaplock modify -vserver vs1 -volume voll -minimum  
-retention-period 10days  
cluster1::> volume snaplock modify -vserver vs1 -volume voll -default  
-retention-period min
```

The following command sets the default retention period for a Compliance volume to infinite:

```
cluster1::> volume snaplock modify -vserver vs1 -volume voll -default  
-retention-period infinite -maximum-retention-period infinite
```

Set the retention time for a file explicitly

You can set the retention time for a file explicitly by modifying its last access time. You can use any suitable command or program over NFS or CIFS to modify the last access time.

About this task

After a file has been committed to WORM, you can extend but not shorten the retention time. The retention time is stored in the `atime` field for the file.



You cannot explicitly set the retention time of a file to `infinite`. That value is only available when you use the default retention period to calculate the retention time.

Steps

1. Use a suitable command or program to modify the last access time for the file whose retention time you want to set.

In a UNIX shell, use the following command to set a retention time of 21 November 2020 6:00 a.m. on a file named `document.txt`:

```
touch -a -t 202011210600 document.txt
```



You can use any suitable command or program to modify the last access time in Windows.

Set the file retention period after an event

Beginning with ONTAP 9.3, you can define how long a file is retained after an event occurs by using the SnapLock *Event Based Retention (EBR)* feature.

What you'll need

- You must be a SnapLock administrator to perform this task.

[Create a SnapLock administrator account](#)

- You must have logged in on a secure connection (SSH, console, or ZAPI).

About this task

The *event retention policy* defines the retention period for the file after the event occurs. The policy can be applied to a single file or all the files in a directory.

- If a file is not a WORM file, it will be committed to the WORM state for the retention period defined in the policy.
- If a file is a WORM file or a WORM appendable file, its retention period will be extended by the retention period defined in the policy.

You can use a Compliance-mode or Enterprise-mode volume.



EBR policies cannot be applied to files under a Legal Hold.

For advanced usage, see [Compliant WORM Storage Using NetApp SnapLock](#).

Using EBR to extend the retention period of already existing WORM files

EBR is convenient when you want to extend the retention period of already existing WORM files. For example, it might be your firm's policy to retain employee W-4 records in unmodified form for three years after the employee changes a withholding election. Another company policy might require that W-4 records be retained for five years after the employee is terminated.

In this situation, you could create an EBR policy with a five-year retention period. After the employee is terminated (the "event"), you would apply the EBR policy to the employee's W-4 record, causing its retention period to be extended. That will usually be easier than extending the retention period manually, particularly when a large number of files is involved.

Steps

1. Create an EBR policy:

```
snaplock event-retention policy create -vserver SVM_name -name policy_name
-retention-period retention_period
```

The following command creates the EBR policy `employee_exit` on `vs1` with a retention period of ten years:

```
cluster1::>snaplock event-retention policy create -vserver vs1 -name
employee_exit -retention-period 10years
```

2. Apply an EBR policy:

```
snaplock event-retention apply -vserver SVM_name -name policy_name -volume
volume_name -path path_name
```


The following command applies the EBR policy `employee_exit` on `vs1` to all the files in the directory `d1`:

```
cluster1::>snaplock event-retention apply -vserver vs1 -name  
employee_exit -volume vol1 -path /d1
```

Create an audit log

You must create a SnapLock-protected audit log before performing a privileged delete or SnapLock volume move. The audit log records the creation and deletion of SnapLock administrator accounts, modifications to the log volume, whether privileged delete is enabled, privileged delete operations, and SnapLock volume move operations.

What you'll need

You must be a cluster administrator to create a SnapLock aggregate.

About this task

You cannot delete an audit log until the log file retention period has elapsed. You cannot modify an audit log even after the retention period has elapsed. This is true for both SnapLock Compliance and Enterprise modes.



In ONTAP 9.4 and earlier, you cannot use a SnapLock Enterprise volume for audit logging. You must use a SnapLock Compliance volume. In ONTAP 9.5 and later, you can use either a SnapLock Enterprise volume or a SnapLock Compliance volume for audit logging. In all cases, the audit log volume must be mounted at the junction path `/snaplock_audit_log`. No other volume can use this junction path.

You can find the SnapLock audit logs in the `/snaplock_log` directory under the root of the audit log volume, in subdirectories named `privdel_log` (privileged delete operations) and `system_log` (everything else). Audit log file names contain the timestamp of the first logged operation, making it easy to search for records by the approximate time that operations were executed.

- You can use the `snaplock log file show` command to view the log files on the audit log volume.
- You can use the `snaplock log file archive` command to archive the current log file and create a new one, which is useful in cases where you need to record audit log information in a separate file.

For more information, see the man pages for the commands.



A data protection volume cannot be used as a SnapLock audit log volume.

Steps

1. Create a SnapLock aggregate.

[Create a SnapLock aggregate](#)

2. On the SVM that you want to configure for audit logging, create a SnapLock volume.

[Create a SnapLock volume](#)

3. Configure the SVM for audit logging:

```
snaplock log create -vserver SVM_name -volume snaplock_volume_name -max-file  
-size size -retention-period default_retention_period
```



The minimum default retention period for audit log files is six months. If the retention period of an affected file is longer than the retention period of the audit log, the retention period of the log inherits the retention period of the file. So, if the retention period for a file deleted using privileged delete is 10 months, and the retention period of the audit log is 8 months, the retention period of the log is extended to 10 months. For more information about retention time and default retention period, see [Set the retention time](#).

The following command configures SVM1 for audit logging using the SnapLock volume logVol. The audit log has a maximum size of 20 GB and is retained for eight months.

```
SVM1::> snaplock log create -vserver SVM1 -volume logVol -max-file-size  
20GB -retention-period 8months
```

4. On the SVM that you configured for audit logging, mount the SnapLock volume at the junction path /snaplock_audit_log.

[Mount a SnapLock volume](#)

Verify SnapLock settings

You can use the `volume file fingerprint start` and `volume file fingerprint dump` commands to view key information about files and volumes, including the file type (regular, WORM, or WORM appendable), the volume expiration date, and so forth.

Steps

1. Generate a file fingerprint:

```
volume file fingerprint start -vserver SVM_name -file file_path
```

```
svm1::> volume file fingerprint start -vserver svm1 -file  
/vol/sle/vol/fl  
File fingerprint operation is queued. Run "volume file fingerprint show  
-session-id 16842791" to view the fingerprint session status.
```

The command generates a session ID that you can use as input to the `volume file fingerprint dump` command.



You can use the `volume file fingerprint show` command with the session ID to monitor the progress of the fingerprint operation. Make sure that the operation has completed before attempting to display the fingerprint.

2. Display the fingerprint for the file:

volume file fingerprint dump -session-id *session_ID*

```
svml:> volume file fingerprint dump -session-id 33619976
Vserver:svml
Session-ID:33619976
Volume:slc_vol
Path:/vol/slc_vol/fl
Data
Fingerprint:MOFJVevxNSJm3C/4Bn5oEEYH51CrudOzZYK4r5Cfylg=Metadata

Fingerprint:8iMjqJXiNcqqXT5XuRhLiEwIrJEihDmwS0hrexnjgmc=Fingerprint
Algorithm:SHA256
    Fingerprint Scope:data-and-metadata
    Fingerprint Start Time:1460612586
    Formatted Fingerprint Start Time:Thu Apr 14 05:43:06 GMT 2016
    Fingerprint Version:3
    **SnapLock License:available**
    Vserver UUID:acf7ae64-00d6-11e6-a027-0050569c55ae
    Volume MSID:2152884007
    Volume DSID:1028
    Hostname:my_host
    Filer ID:5f18eda2-00b0-11e6-914e-6fb45e537b8d
    Volume Containing Aggregate:slc_aggr1
    Aggregate ID:c84634aa-c757-4b98-8f07-eeef32565f67
    **SnapLock System ComplianceClock:1460610635
    Formatted SnapLock System ComplianceClock:Thu Apr 14 05:10:35
GMT 2016
    Volume SnapLock Type:compliance
    Volume ComplianceClock:1460610635
    Formatted Volume ComplianceClock:Thu Apr 14 05:10:35 GMT 2016
    Volume Expiry Date:1465880998**
    Is Volume Expiry Date Wraparound:false
    Formatted Volume Expiry Date:Tue Jun 14 05:09:58 GMT 2016
    Filesystem ID:1028
    File ID:96
    File Type:worm
    File Size:1048576
    Creation Time:1460612515
    Formatted Creation Time:Thu Apr 14 05:41:55 GMT 2016
    Modification Time:1460612515
    Formatted Modification Time:Thu Apr 14 05:41:55 GMT 2016
    Changed Time:1460610598
    Is Changed Time Wraparound:false
    Formatted Changed Time:Thu Apr 14 05:09:58 GMT 2016
    Retention Time:1465880998
    Is Retention Time Wraparound:false
```

```
Formatted Retention Time:Tue Jun 14 05:09:58 GMT 2016
Access Time:-
Formatted Access Time:-
Owner ID:0
Group ID:0
Owner SID:-
Fingerprint End Time:1460612586
Formatted Fingerprint End Time:Thu Apr 14 05:43:06 GMT 2016
```

Manage WORM files

Manage WORM files

You can manage WORM files in the following ways:

- [Commit files to WORM](#)
- [Commit Snapshot copies to WORM on a vault destination](#)
- [Mirror WORM files for disaster recovery](#)
- [Retain WORM files during litigation](#)
- [Delete WORM files](#)

Commit files to WORM

You can commit files to WORM (write once, read many) either manually or by committing them automatically. You can also create WORM appendable files.

Commit files to WORM manually

You commit a file to WORM manually by making the file read-only. You can use any suitable command or program over NFS or CIFS to change the read-write attribute of a file to read-only. You might choose to manually commit files if you want to ensure an application has finished writing to a file so that the file isn't committed prematurely or if there are scaling issues for the autocommit scanner because of a high number of volumes.

What you'll need

- The file you want to commit must reside on a SnapLock volume.
- The file must be writable.

About this task

The volume ComplianceClock time is written to the `ctime` field of the file when the command or program is executed. The ComplianceClock time determines when the retention time for the file has been reached.

Steps

1. Use a suitable command or program to change the read-write attribute of a file to read-only.

In a UNIX shell, use the following command to make a file named `document.txt` read-only:

```
chmod -w document.txt
```

In a Windows shell, use the following command to make a file named `document.txt` read-only:

```
attrib +r document.txt
```

Commit files to WORM automatically

The SnapLock autocommit feature enables you to commit files to WORM automatically. The autocommit feature commits a file to WORM state on a SnapLock volume if the file did not change for the autocommit-period duration. The autocommit feature is disabled by default.

What you'll need

- The files you want to autocommit must reside on a SnapLock volume.
- The SnapLock volume must be online.
- The SnapLock volume must be a read-write volume.



The SnapLock autocommit feature scans through all of the files in the volume and commits a file if it meets the autocommit requirement. There might be a time interval between when the file is ready for autocommit and when it is actually committed by the SnapLock autocommit scanner. However, the file is still protected from modifications and deletion by the file system as soon as it is eligible for autocommit.

About this task

The *autocommit period* specifies the amount of time that files must remain unchanged before they are autocommitted. Changing a file before the autocommit period has elapsed restarts the autocommit period for the file.

The following table shows the possible values for the autocommit period:

Value	Unit	Notes
none	-	The default.
5 - 5256000	minutes	-
1 - 87600	hours	-
1 - 3650	days	-
1 - 120	months	-
1 - 10	years	-



The minimum value is 5 minutes and the maximum value is 10 years.

Steps

1. Autocommit files on a SnapLock volume to WORM:

```
volume snaplock modify -vserver SVM_name -volume volume_name -autocommit  
-period autocommit_period
```

For a complete list of options, see the man page for the command.

The following command autocommits the files on volume `vol1` of SVM `vs1`, as long as the files remain unchanged for 5 hours:

```
cluster1::>volume snaplock modify -vserver vs1 -volume vol1 -autocommit  
-period 5hours
```

Create a WORM appendable file

A WORM appendable file retains data written incrementally, like log entries. You can use any suitable command or program to create a WORM appendable file, or you can use the SnapLock *volume append mode* feature to create WORM appendable files by default.

Use a command or program to create a WORM appendable file

You can use any suitable command or program over NFS or CIFS to create a WORM appendable file. A WORM appendable file retains data written incrementally, like log entries. Data is appended to the file in 256 KB chunks. As each chunk is written, the previous chunk becomes WORM-protected. You cannot delete the file until the retention period has elapsed.

What you'll need

The WORM appendable file must reside on a SnapLock volume.

About this task

Data does not have to be written sequentially to the active 256 KB chunk. When data is written to byte $n \times 256\text{KB} + 1$ of the file, the previous 256 KB segment becomes WORM-protected.

Steps

1. Use a suitable command or program to create a zero-length file with the desired retention time.

In a UNIX shell, use the following command to set a retention time of 21 November 2020 6:00 a.m. on a zero-length file named `document.txt`:

```
touch -a -t 202011210600 document.txt
```

2. Use a suitable command or program to change the read-write attribute of the file to read-only.

In a UNIX shell, use the following command to make a file named `document.txt` read-only:

```
chmod 444 document.txt
```

3. Use a suitable command or program to change the read-write attribute of the file back to writable.



This step is not deemed a compliance risk because there is no data in the file.

In a UNIX shell, use the following command to make a file named `document.txt` writable:

```
chmod 777 document.txt
```

4. Use a suitable command or program to start writing data to the file.

In a UNIX shell, use the following command to write data to `document.txt`:

```
echo test data >> document.txt
```



Change the file permissions back to read-only when you no longer need to append data to the file.

Use volume append mode to create WORM appendable files

Beginning with ONTAP 9.3, you can use the SnapLock *volume append mode* (VAM) feature to create WORM appendable files by default. A WORM appendable file retains data written incrementally, like log entries. Data is appended to the file in 256 KB chunks. As each chunk is written, the previous chunk becomes WORM-protected. You cannot delete the file until the retention period has elapsed.

What you'll need

- The WORM appendable file must reside on a SnapLock volume.
- The SnapLock volume must be unmounted and empty of Snapshot copies and user-created files.

About this task

Data does not have to be written sequentially to the active 256 KB chunk. When data is written to byte $n \times 256\text{KB} + 1$ of the file, the previous 256 KB segment becomes WORM-protected.

If you specify an autocommit period for the volume, WORM appendable files that are not modified for a period greater than the autocommit period are committed to WORM.



VAM is not supported on SnapLock audit log volumes.

Steps

1. Enable VAM:

```
volume snaplock modify -vserver SVM_name -volume volume_name -is-volume-append  
-mode-enabled true|false
```

For a complete list of options, see the man page for the command.

The following command enables VAM on volume `vol1` of SVM`vs1`:

```
cluster1::>volume snaplock modify -vserver vs1 -volume vol1 -is-volume  
-append-mode-enabled true
```

2. Use a suitable command or program to create files with write permissions.

The files are WORM-appendable by default.

Commit Snapshot copies to WORM on a vault destination

You can use SnapLock for SnapVault to WORM-protect Snapshot copies on secondary storage. You perform all of the basic SnapLock tasks on the SnapVault destination. The destination volume is automatically mounted read-only, so there is no need to explicitly commit the Snapshot copies to WORM; therefore, creating scheduled Snapshot copies on the destination volume using SnapMirror policies is not supported.

Before you begin

- The source cluster must be running ONTAP 8.2.2 or later.
- The source and destination aggregates must be 64-bit.
- The source volume cannot be a SnapLock volume.
- The source and destination volumes must be created in peered clusters with peered SVMs.

For more information, see [Cluster Peering](#).

- If volume autogrow is disabled, the free space on the destination volume must be at least five percent more than the used space on the source volume.

About this task

The source volume can use NetApp or non-NetApp storage. For non-NetApp storage, you must use FlexArray Virtualization.



You cannot rename a Snapshot copy that is committed to the WORM state.

You can clone SnapLock volumes, but you cannot clone files on a SnapLock volume.



LUNs are not supported on SnapLock volumes. Although it is possible to move LUNs onto a SnapLock volume using legacy technology, this is not a supported operation, nor is any other operation involving LUNs on a SnapLock volume. Beginning with ONTAP 9.9.0, LUNs on a SnapLock volume are supported in SnapLock *only* for SnapVault relationships where a Snapshot copy of a non-SnapLock source volume is replicated and locked on a SnapLock destination. These Snapshot copies can contain LUNs.

Beginning with ONTAP 9.13.1, you can instantaneously restore a locked Snapshot copy on the destination SnapLock volume of a SnapLock for SnapVault relationship by creating a FlexClone with the `snaplock-type` option set to “non-snaplock” and specifying the Snapshot copy as the “parent-snapshot” when executing the volume clone creation operation. Learn more about [creating a FlexClone volume with a SnapLock type](#).

For MetroCluster configurations, you should be aware of the following:

- You can create a SnapVault relationship only between sync-source SVMs, not between a sync-source SVM and a sync-destination SVM.
- You can create a SnapVault relationship from a volume on a sync-source SVM to a data-serving SVM.
- You can create a SnapVault relationship from a volume on a data-serving SVM to a DP volume on a sync-source SVM.

The following illustration shows the procedure for initializing a SnapVault relationship:

Steps

1. Identify the destination cluster.
2. On the destination cluster, install the SnapLock license, initialize the ComplianceClock, and, if you are using an ONTAP release earlier than 9.10.1, create a SnapLock aggregate, as described in [SnapLock workflow](#).
3. On the destination cluster, create a SnapLock destination volume of type DP that is either the same or greater in size than the source volume:

```
volume create -vserver SVM_name -volume volume_name -aggregate aggregate_name  
-snaplock-type compliance|enterprise -type DP -size size
```



Beginning with ONTAP 9.10.1, SnapLock and non-SnapLock volumes can exist on the same aggregate; therefore, you are no longer required to create a separate SnapLock aggregate if you are using ONTAP 9.10.1. You use the volume `-snaplock-type` option to specify a Compliance or Enterprise SnapLock volume type. In ONTAP releases earlier than ONTAP 9.10.1, the SnapLock mode, Compliance or Enterprise, is inherited from the aggregate. Version-flexible destination volumes are not supported. The language setting of the destination volume must match the language setting of the source volume.

The following command creates a 2 GB SnapLock Compliance volume named `dstvolB` in SVM2 on the aggregate `node01_aggr`:

```
cluster2::> volume create -vserver SVM2 -volume dstvolB -aggregate  
node01_aggr -snaplock-type compliance -type DP -size 2GB
```

4. On the destination cluster, set the default retention period, as described in [Set the default retention period](#).



A SnapLock volume that is a vault destination has a default retention period assigned to it. The value for this period is initially set to a minimum of 0 years for SnapLock Enterprise volumes and a maximum of 30 years for SnapLock Compliance volumes. Each NetApp Snapshot copy is committed with this default retention period at first. The retention period can be extended later, if needed. For more information, see [Set retention time overview](#).

5. [Create a new replication relationship](#) between the non-SnapLock source and the new SnapLock destination you created in Step 3.

This example creates a new SnapMirror relationship with destination SnapLock volume `dstvolB` using a policy of `XDPEndpoint` to vault Snapshot copies labeled daily and weekly on an hourly schedule:

```
cluster2::> snapmirror create -source-path SVM1:srcvolA -destination  
-path SVM2:dstvolB -vserver SVM2 -policy XDPDefault -schedule hourly
```



Create a [custom replication policy](#) or a [custom schedule](#) if the available defaults are not suitable.

6. On the destination SVM, initialize the SnapVault relationship created in Step 5:

```
snapmirror initialize -destination-path destination_path
```

The following command initializes the relationship between the source volume `srcvolA` on SVM1 and the destination volume `dstvolB` on SVM2:

```
cluster2::> snapmirror initialize -destination-path SVM2:dstvolB
```

7. After the relationship is initialized and idle, use the `snapshot show` command on the destination to verify the SnapLock expiry time applied to the replicated Snapshot copies.

This example lists the Snapshot copies on volume `dstvolB` that have the SnapMirror label and the SnapLock expiration date:

```
cluster2::> snapshot show -vserver SVM2 -volume dstvolB -fields  
snapmirror-label, snaplock-expiry-time
```

Related information

[Cluster and SVM peering](#)

[Volume backup using SnapVault](#)

Mirror WORM files for disaster recovery

You can use SnapMirror to replicate WORM files to another geographic location for disaster recovery and other purposes. Both the source volume and destination volume must be configured for SnapLock, and both volumes must have the same SnapLock mode, Compliance or Enterprise. All key SnapLock properties of the volume and files are replicated.

Prerequisites

The source and destination volumes must be created in peered clusters with peered SVMs. For more information, see [Cluster and SVM peering](#).

About this task

- Beginning with ONTAP 9.5, you can replicate WORM files with the XDP (extended data protection) type SnapMirror relationship rather than the DP (data protection) type relationship. XDP mode is ONTAP version-independent, and is able to differentiate files stored in the same block, making it much easier to resync replicated Compliance-mode volumes. For information on how to convert an existing DP-type

relationship to an XDP-type relationship, see [Data Protection](#).

- A resync operation on a DP type SnapMirror relationship fails for a Compliance-mode volume if SnapLock determines that it will result in a loss of data. If a resync operation fails, you can use the `volume clone create` command to make a clone of the destination volume. You can then resync the source volume with the clone.
- A SnapMirror relationship of type XDP between SnapLock compliant volumes supports a resync after a break even if data on the destination has diverged from the source post the break.

On a resync, when data divergence is detected between the source the destination beyond the common snapshot, a new snapshot is cut on the destination to capture this divergence. The new snapshot and the common snapshot are both locked with a retention time as follows:

- The volume expiry time of the destination
- If the volume expiry time is in the past or has not been set, then the snapshot is locked for a period of 30 days
- If the destination has legal-holds, the actual volume expiry period is masked and shows up as 'indefinite', however the snapshot is locked for the duration of the actual volume expiry period.

If the destination volume has an expiry period that is later than the source, the destination expiry period is retained and will not be overwritten by the expiry period of the source volume post the resync.

If the destination has legal-holds placed on it that differ from the source, a resync is not allowed. The source and destination must have identical legal-holds or all legal-holds on the destination must be released before a resync is attempted.

A locked Snapshot copy on the destination volume created to capture the divergent data can be copied to the source using the CLI by running the `snapmirror update -s snapshot` command. The snapshot once copied will continue to be locked at the source as well.

- SVM data protection relationships are not supported.
- Load-sharing data protection relationships are not supported.

The following illustration shows the procedure for initializing a SnapMirror relationship:

System Manager

Beginning with ONTAP 9.12.1, you can use System Manager to set up SnapMirror replication of WORM files.

Steps

1. Navigate to **Storage > Volumes**.
2. Click **Show/Hide** and select **SnapLock Type** to display the column in the **Volumes** window.
3. Locate a SnapLock volume.
4. Click  and select **Protect**.
5. Choose the destination cluster and the destination storage VM.
6. Click **More Options**.
7. Select **Show legacy policies** and select **DPDefault (legacy)**.
8. In the **Destination Configuration details** section, select **Override transfer schedule** and select **hourly**.
9. Click **Save**.
10. To the left of the source volume name, click the arrow to expand the volume details, and on the right side of the page, review the remote SnapMirror protection details.
11. On the remote cluster, navigate to **Protection Relationships**.
12. Locate the relationship and click the destination volume name to view the relationship details.
13. Verify that the destination volume SnapLock type and other SnapLock information.

CLI

1. Identify the destination cluster.
2. On the destination cluster, install the SnapLock license, initialize the ComplianceClock, and, if you are using an ONTAP release earlier than 9.10.1, create a SnapLock aggregate.
3. On the destination cluster, create a SnapLock destination volume of type **DP** that is either the same size as or greater in size than the source volume:

```
volume create -vserver SVM_name -volume volume_name -aggregate  
aggregate_name -snaplock-type compliance|enterprise -type DP -size size
```



Beginning with ONTAP 9.10.1, SnapLock and non-SnapLock volumes can exist on the same aggregate; therefore, you are no longer required to create a separate SnapLock aggregate if you are using ONTAP 9.10.1. You use the volume `-snaplock-type` option to specify a Compliance or Enterprise SnapLock volume type. In ONTAP releases earlier than ONTAP 9.10.1, the SnapLock mode—Compliance or Enterprise—is inherited from the aggregate. Version-flexible destination volumes are not supported. The language setting of the destination volume must match the language setting of the source volume.

The following command creates a 2 GB SnapLock Compliance volume named `dstvolB` in SVM2 on the aggregate `node01_aggr`:

```
cluster2::> volume create -vserver SVM2 -volume dstvolB -aggregate  
node01_aggr -snaplock-type compliance -type DP -size 2GB
```

4. On the destination SVM, create a SnapMirror policy:

```
snapmirror policy create -vserver SVM_name -policy policy_name
```

The following command creates the SVM-wide policy SVM1-mirror:

```
SVM2::> snapmirror policy create -vserver SVM2 -policy SVM1-mirror
```

5. On the destination SVM, create a SnapMirror schedule:

```
job schedule cron create -name schedule_name -dayofweek day_of_week -hour  
hour -minute minute
```

The following command creates a SnapMirror schedule named weekendcron:

```
SVM2::> job schedule cron create -name weekendcron -dayofweek  
"Saturday, Sunday" -hour 3 -minute 0
```

6. On the destination SVM, create a SnapMirror relationship:

```
snapmirror create -source-path source_path -destination-path  
destination_path -type XDP|DP -policy policy_name -schedule schedule_name
```

The following command creates a SnapMirror relationship between the source volume `srcvolA` on SVM1 and the destination volume `dstvolB` on SVM2, and assigns the policy `SVM1-mirror` and the schedule `weekendcron`:

```
SVM2::> snapmirror create -source-path SVM1:srcvolA -destination  
-path SVM2:dstvolB -type XDP -policy SVM1-mirror -schedule  
weekendcron
```



The XDP type is available in ONTAP 9.5 and later. You must use the DP type in ONTAP 9.4 and earlier.

7. On the destination SVM, initialize the SnapMirror relationship:

```
snapmirror initialize -destination-path destination_path
```

The initialization process performs a *baseline transfer* to the destination volume. SnapMirror makes a Snapshot copy of the source volume, then transfers the copy and all the data blocks that it references to the destination volume. It also transfers any other Snapshot copies on the source volume to the destination volume.

The following command initializes the relationship between the source volume `srcvolA` on SVM1 and the destination volume `dstvolB` on SVM2:

```
SVM2::> snapmirror initialize -destination-path SVM2:dstvolB
```

Related information

[Cluster and SVM peering](#)

[Volume disaster recovery preparation](#)

[Data protection](#)

Retain WORM files during litigation using Legal Hold

Beginning with ONTAP 9.3, you can retain Compliance-mode WORM files for the duration of a litigation by using the *Legal Hold* feature.

What you'll need

- You must be a SnapLock administrator to perform this task.

[Create a SnapLock administrator account](#)

- You must have logged in on a secure connection (SSH, console, or ZAPI).

About this task

A file under a Legal Hold behaves like a WORM file with an indefinite retention period. It is your responsibility to specify when the Legal Hold period ends.

The number of files you can place under a Legal Hold depends on the space available on the volume.

Steps

1. Start a Legal Hold:

```
snaplock legal-hold begin -litigation-name litigation_name -volume volume_name -path path_name
```

The following command starts a Legal Hold for all the files in `vol1`:

```
cluster1::> snaplock legal-hold begin -litigation-name litigation1 -volume vol1 -path /
```

2. End a Legal Hold:

```
snaplock legal-hold end -litigation-name litigation_name -volume volume_name -path path_name
```

The following command ends a Legal Hold for all the files in `vol1`:

```
cluster1::>snaplock legal-hold end -litigation-name litigation1 -volume  
voll1 -path /
```

Delete WORM files overview

You can delete Enterprise-mode WORM files during the retention period using the privileged delete feature.

Before you can use this feature, you must create a SnapLock administrator account and then using the account, enable the feature.

Create a SnapLock administrator account

You must have SnapLock administrator privileges to perform a privileged delete. These privileges are defined in the vsadmin-snaplock role. If you have not already been assigned that role, you can ask your cluster administrator to create an SVM administrator account with the SnapLock administrator role.

What you'll need

- You must be a cluster administrator to perform this task.
- You must have logged in on a secure connection (SSH, console, or ZAPI).

Steps

1. Create an SVM administrator account with the SnapLock administrator role:

```
security login create -vserver SVM_name -user-or-group-name user_or_group_name  
-application application -authmethod authentication_method -role role -comment  
comment
```

The following command enables the SVM administrator account SnapLockAdmin with the predefined vsadmin-snaplock role to access SVM1 using a password:

```
cluster1::> security login create -vserver SVM1 -user-or-group-name  
SnapLockAdmin -application ssh -authmethod password -role vsadmin-  
snaplock
```

Enable the privileged delete feature

You must explicitly enable the privileged delete feature on the Enterprise volume that contains the WORM files you want to delete.

About this task

The value of the `-privileged-delete` option determines whether privileged delete is enabled. Possible values are enabled, disabled, and permanently-disabled.



permanently-disabled is the terminal state. You cannot enable privileged delete on the volume after you set the state to permanently-disabled.

Steps

1. Enable privileged delete for a SnapLock Enterprise volume:

```
volume snaplock modify -vserver SVM_name -volume volume_name -privileged  
-delete disabled|enabled|permanently-disabled
```

The following command enables the privileged delete feature for the Enterprise volume dataVol on SVM1:

```
SVM1::> volume snaplock modify -vserver SVM1 -volume dataVol -privileged  
-delete enabled
```

Delete Enterprise-mode WORM files

You can use the privileged delete feature to delete Enterprise-mode WORM files during the retention period.

What you'll need

- You must be a SnapLock administrator to perform this task.
- You must have created a SnapLock audit log and enabled the privileged delete feature on the Enterprise volume.

About this task

You cannot use a privileged delete operation to delete an expired WORM file. You can use the `volume file retention show` command to view the retention time of the WORM file that you want to delete. For more information, see the man page for the command.

Step

1. Delete a WORM file on an Enterprise volume:

```
volume file privileged-delete -vserver SVM_name -file file_path
```

The following command deletes the file `/vol/dataVol/f1` on the SVM `SVM1`:

```
SVM1::> volume file privileged-delete -file /vol/dataVol/f1
```

Move a SnapLock volume

Beginning with ONTAP 9.8, you can move a SnapLock volume to a destination aggregate of the same type, either Enterprise to Enterprise, or Compliance to Compliance. You must be assigned the SnapLock security role to move a SnapLock volume.

Create a SnapLock security administrator account

You must have SnapLock security administrator privileges to perform a SnapLock volume move. This privilege is granted to you with the `snaplock` role, introduced in ONTAP 9.8. If you have not already been assigned that role, you can ask your cluster administrator to create a SnapLock security user with this SnapLock security role.

What you'll need

- You must be a cluster administrator to perform this task.
- You must have logged in on a secure connection (SSH, console, or ZAPI).

About this task

The snaplock role is associated with the admin SVM, unlike the vsadmin-snaplock role, which is associated with the data SVM.

Step

1. Create an SVM administrator account with the SnapLock administrator role:

```
security login create -vserver SVM_name -user-or-group-name user_or_group_name  
-application application -authmethod authentication_method -role role -comment  
comment
```

The following command enables the SVM administrator account SnapLockAdmin with the predefined snaplock role to access admin SVM cluster1 using a password:

```
cluster1::> security login create -vserver cluster1 -user-or-group-name  
SnapLockAdmin -application ssh -authmethod password -role snaplock
```

Move a SnapLock volume

You can use the `volume move` command to move a SnapLock volume to a destination aggregate.

What you'll need

- You must have created a SnapLock-protected audit log before performing SnapLock volume move.

[Create an audit log.](#)

- If you are using a version of ONTAP earlier than ONTAP 9.10.1, the destination aggregate must be the same SnapLock type as the SnapLock volume you want to move; either Compliance to Compliance or Enterprise to Enterprise. Beginning with ONTAP 9.10.1, this restriction is removed and an aggregate can include both Compliance and Enterprise SnapLock volumes, as well as non-SnapLock volumes.
- You must be a user with the SnapLock security role.

Steps

1. Using a secure connection, log in to the ONTAP cluster management LIF:

```
ssh snaplock_user@cluster_mgmt_ip
```

2. Move a SnapLock volume:

```
volume move start -vserver SVM_name -volume SnapLock_volume_name -destination  
-aggregate destination_aggregate_name
```

3. Check the status of the volume move operation:

```
volume move show -volume SnapLock_volume_name -vserver SVM_name -fields  
volume,phase,vserver
```

Lock a Snapshot copy for protection against ransomware attacks

Beginning with ONTAP 9.12.1, you can lock a Snapshot copy on a non-SnapLock volume to provide protection from ransomware attacks. Locking Snapshot copies ensures that they can't be deleted accidentally or maliciously.

You use the SnapLock compliance clock feature to lock Snapshot copies for a specified period so that they cannot be deleted until the expiration time is reached. Locking Snapshot copies makes them tamperproof, protecting them from ransomware threats. You can use locked Snapshot copies to recover data if a volume is compromised by a ransomware attack.

Tamperproof Snapshot copy requirements and considerations

- If you are using the ONTAP CLI, all nodes in the cluster must be running ONTAP 9.12.1 or later. If you are using System Manager, all nodes must be running ONTAP 9.13.1 or later.
- The SnapLock license must be installed on the cluster.

For details, see [Installing the SnapLock license](#).

- The compliance clock on the cluster must be initialized.

For details, see [Initialize the Compliance Clock](#).

- When Snapshot locking is enabled on a volume, you can upgrade the clusters to a version of ONTAP later than ONTAP 9.12.1; however, you cannot revert to an earlier version of ONTAP until all locked Snapshot copies have reached their expiration date and are deleted and Snapshot copy locking is disabled.
- When a Snapshot is locked, the volume expiry time is set to the expiry time of the Snapshot copy. If more than one Snapshot copy is locked, the volume expiry time reflects the largest expiry time among all Snapshot copies.
- The retention period for locked Snapshot copies takes precedence over the Snapshot copy keep count, which means the keep count limit is not honored if the Snapshot copy retention period for locked Snapshot copies has not expired.
- In a SnapMirror relationship, you can set a retention period on a mirror-vault policy rule, and the retention period is applied for Snapshot copies replicated to the destination if the destination volume has Snapshot copy locking enabled. The retention period takes precedence over keep count; for example, Snapshot copies that have not passed their expiry will be retained even if the keep count is exceeded.
- You can rename a Snapshot copy on a non-SnapLock volume. Snapshot rename operations on the primary volume of a SnapMirror relationship are reflected on the secondary volume only if the policy is MirrorAllSnapshots. For other policy types, the renamed Snapshot copy is not propagated during updates.
- If you are using the ONTAP CLI, you can restore a locked Snapshot copy with the `volume snapshot restore` command only if the locked Snapshot copy is the most recent. If there are any unexpired Snapshot copies later than the one being restored, the Snapshot copy restore operation fails.

Features supported with tamperproof Snapshot copies

- FlexGroup volumes

Snapshot copy locking is supported on FlexGroup volumes. Snapshot locking occurs only on the root constituent Snapshot copy. Deleting the FlexGroup volume is allowed only if the root constituent expiration time has passed.

- FlexVol to FlexGroup conversion

You can convert a FlexVol volume with locked Snapshot copies to a FlexGroup volume. Snapshot copies remain locked after the conversion.

- Volume clone and file clone

You can create volume clones and file clones from a locked Snapshot copy.

Unsupported features

The following features currently are not supported with tamperproof Snapshot copies:

- Consistency groups
- FabricPool
- FlexCache volumes
- SMtape
- SnapCenter
- SnapMirror Business Continuity (SM-BC)
- SnapMirror Synchronous
- SVM data mobility

Enable Snapshot copy locking when creating a volume

Beginning with ONTAP 9.12.1, you can enable Snapshot copy locking when you create a new volume or when you modify an existing volume by using the `-snapshot-locking-enabled` option with the `volume create` and `volume modify` commands in the CLI. Beginning with ONTAP 9.13.1, you can use System Manager to enable Snapshot copy locking.